

1 A sphere is projected horizontally. The sphere is photographed at intervals of 0.10 s. The images of the sphere are shown against a grid in Fig 1.1. Air resistance is negligible.

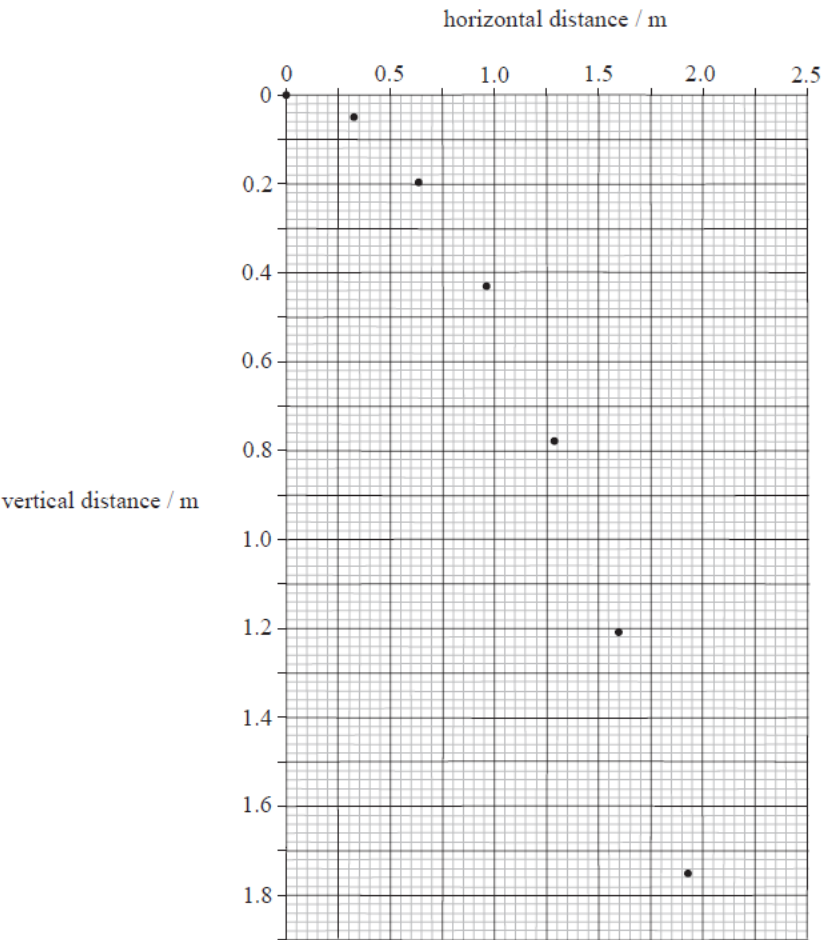


Fig 1.1

(a) Use data from the diagram to determine the acceleration of free fall.

acceleration = m s⁻² [2]

$g = 2s/t^2$
 $s = 1.75 \text{ m}, t = 0.6 \text{ s}$ C1
 $g = 9.7 \text{ m s}^{-2}$ A1

(b) Explain how your choice of the data point from Fig 1.1 helps to improve the reliability of your calculation in **(a)**.

[2]

		% uncertainty = absolute uncertainty/ data point. M1 Hence the larger the data point, the smaller the % uncertainty since the absolute uncertainty is fixed. Hence more reliable. A1	
	(c)	Determine the speed of the sphere 1.2 s after release.	
		speed = m s ⁻¹	[4]
		horizontal speed = 3.2 m s ⁻¹ C1 use of $v = gt$ or $s = \frac{1}{2}gt^2$ vertical speed = 11.6 or 11.7 m s ⁻¹ C1 use of Pythagoras' theorem; M1 speed = 12 m s ⁻¹ A1	
	(d)	On the grid, draw the path of the sphere assuming air resistance is not negligible.	[2]
		line always to left of spheres; B1 becoming more vertical/ steeper; B1	
			[Total: 10]

3	(a)	A prototype space shuttle attached to a rocket stands vertically on its launching pad. The total
---	-----	--